



King Abdulaziz University
College of Engineering
Electrical and Computer Engineering



EE 424 (Antenna)

Assignment L13

(Yagi-Uda Antenna)

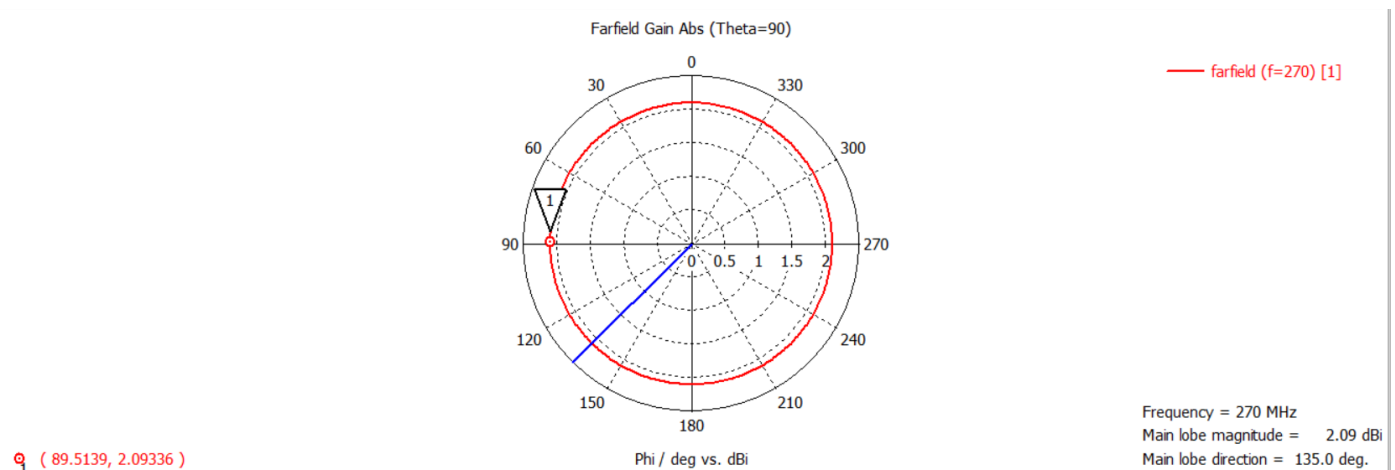
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Requirement: using my previous dipole Antenna, and studying the effects of adding the following parasitic elements on the antenna:

- 1-Adding a reflector element to antenna
- 2-Adding a reflector and 4 directors

Strating with my previous dipole antenna the performance was:



Dipole radiation pattern

The gain was 2.09 dBi.

Then using previous dipole calculations and formulas from L13 to calculate the length of reflector and directors:

$$\text{Lamda} = c/f = 1.11\text{m}$$

Dipole length = $\lambda \cdot 0.47 = 0.522 \text{ m}$

Arm length = 0.261m

Dipole radius = 0.522m

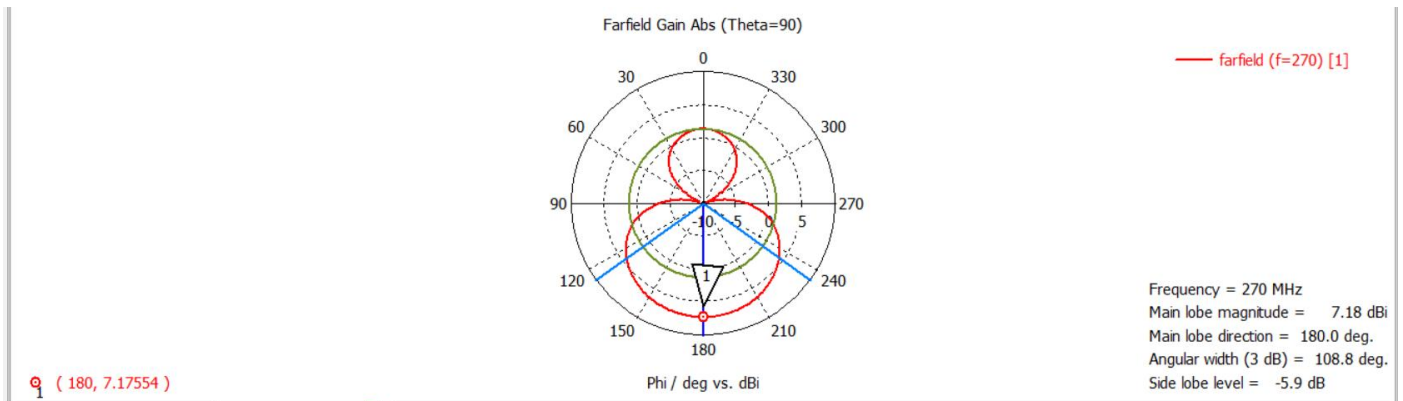
$$G_{ab} = 5.22\text{mm}$$

Reflectors length = $\lambda \cdot 0.482 = 0.535 \text{ m}$

Director length = $\lambda \cdot 0.45 = 0.499\text{m}$

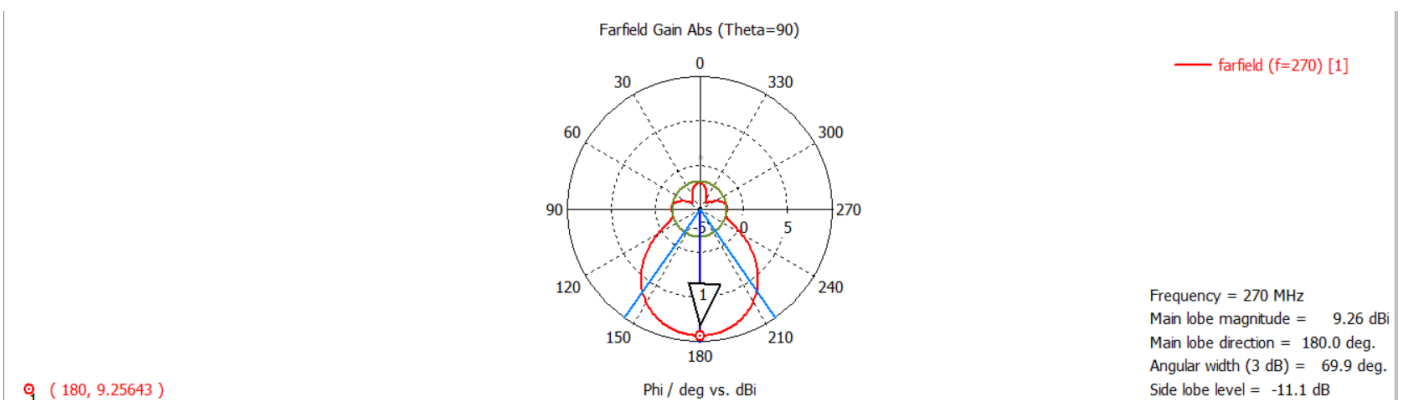
$$\text{Spacing gap} = \lambda \cdot 0.2 = 0.222\text{m}$$

Now after we added the reflector the gain shows improving from 2.09 dBi to 7.18dBi by just adding reflector and that because of focusing the Beam to one direction:



Dipole with reflector radiation pattern

After that we added 4 directors as required and shows a high improving in gain to 9.26dBi due the directivity created by the directors elements and we can see the pattern got lower back and side lobes:



Dipole with reflector and 4 directors radiation

References:

Lecture 13 formulas

Lecture 5 dipole design